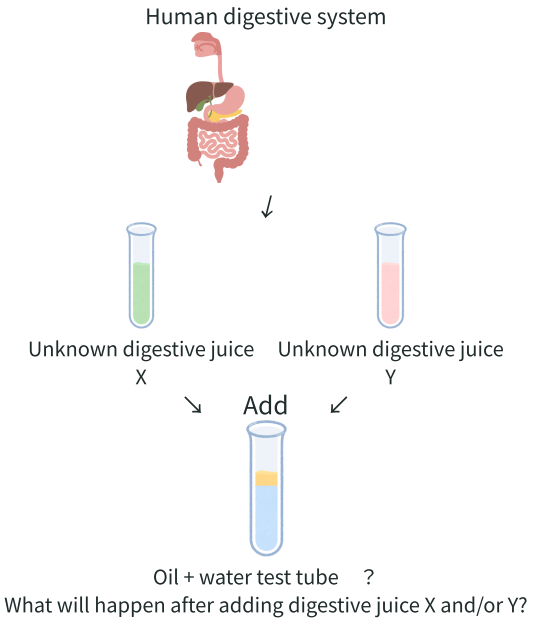
Name: _____ Class and student number: _____

Unknown Digestive Juices X and Y Their effects on lipids



Research background

Lipids in our food have large molecular structures. They cannot pass directly through the cells lining the digestive tract (消化道) to be absorbed.

Research task

How do digestive juices X and Y affect the appearance of an oil–water mixture?

Researchers have collected digestive juices X and Y from the digestive tract (消化道). Design a fair test in this virtual lab to discover their functions.

Your first observation

Describe only what you can see in the oil + water test tube (do not yet guess what X and Y are or what the results will be).

01 • Write an idea to test

Complete your prediction

If I add （Digestive juice X ／ Digestive juice X + digestive juice Y ／ Digestive juice Y） to the oil and water, the mixture in the test tube will （become clear ／ become cloudy ／ show no obvious change）.

Circle one answer in each set of brackets in the sentence.

My reason (required; there is no single correct answer)

02 • Fair test design

You may choose more than one.

Independent variable (the factor changed on purpose)	Dependent variable (the result you measure or observe)	Controlled variables (the factors kept the same)
☐ Total volume of oil and water	☐ Total volume of oil and water	☐ Total volume of oil and water
☐ Reaction temperature	☐ Reaction temperature	☐ Reaction temperature
☐ Reaction time	☐ Reaction time	☐ Reaction time
☐ Total volume of liquid added	☐ Total volume of liquid added	☐ Total volume of liquid added
☐ Method of shaking	☐ Method of shaking	☐ Method of shaking
☐ Appearance of the mixture	☐ Appearance of the mixture	☐ Appearance of the mixture
☐ Combination of digestive juices	☐ Combination of digestive juices	☐ Combination of digestive juices

03 • Conditions for a fair test

Which conditions does this fair test need?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all the conditions that apply. You may choose more than one.

You may choose more than one.

- ☐ Observe all setups after the same reaction time.
- ☐ The X + Y group can have more liquid in total to compare it with the other groups.
- ☐ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.
- ☐ Use the same method of shaking for every setup.

04 • Control group

Does this experiment need a control group? If so, how would you design the setups to compare?

(Hint: A control group provides a starting point for comparison. It helps you check that changes are caused by the independent variable, rather than other outside factors. The control group therefore does not receive the treatment being tested.)

The teacher saves this original design before giving out Part B. Keep a copy to refer to during reflection.

05 • Experiment setup design

Draw and label your setup

Use your experimental variables to help you draw and label your setup.

(Hint: How many setups do you need? Which groups will you test and which is the control group?)


Oil–water test tubes ×4


Digestive juice X ×1


Digestive juice Y ×1


Droppers ×4


37°C water bath ×1

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same.

The teacher saves this original design before giving out Part B. Keep a copy to refer to during reflection.